

Project Title: Retail Sales Data Analysis using Python

This project analyses retail sales data using Python. The dataset was cleaned by handling missing values and duplicate records. Exploratory Data Analysis (EDA) was performed using Pandas and Matplotlib to identify monthly sales trends, top-selling products, category-wise revenue, and region-wise performance. The project helps demonstrate basic data analysis techniques and business decision-making using data visualization.

Objective

- Clean the raw sales dataset.
- Perform Exploratory Data Analysis (EDA).
- Identify sales trends.
- Find best-selling products.
- Compare category-wise revenue.
- Analyse region-wise sales.
- Generate business insights

Tools & Technologies-

Programming Language	Python
Data Cleaning & Analysis	Pandas
Numerical Operations	Numpy
Data Visualization	Matplotlib
Development Environment	vscode

Working Flow –

Data Generation → Data Cleaning → Missing Value Handling → Duplicate Removal → Feature Engineering → EDA → Visualization → Business Insights

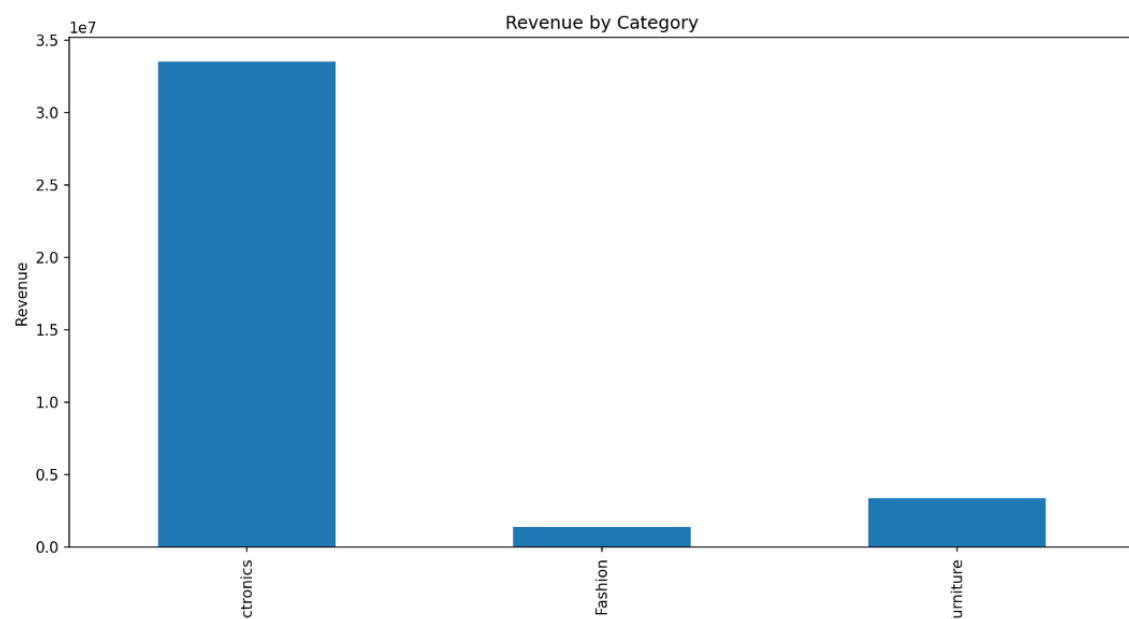
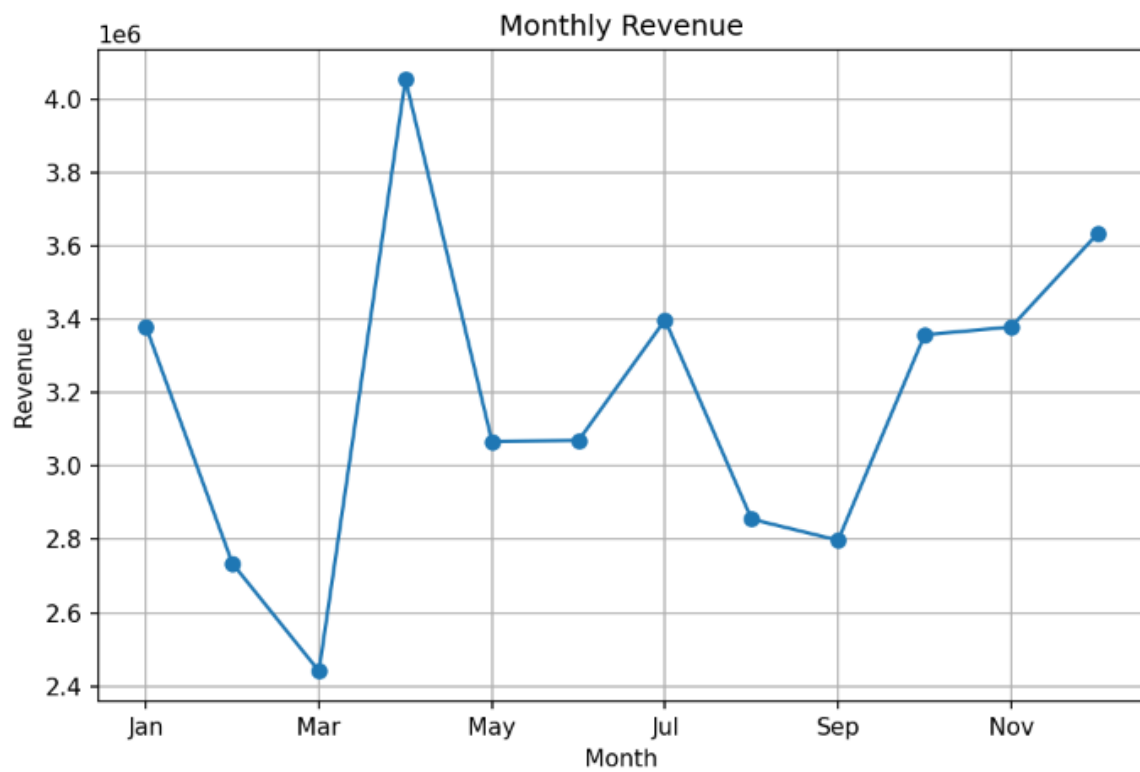
Results and graphs

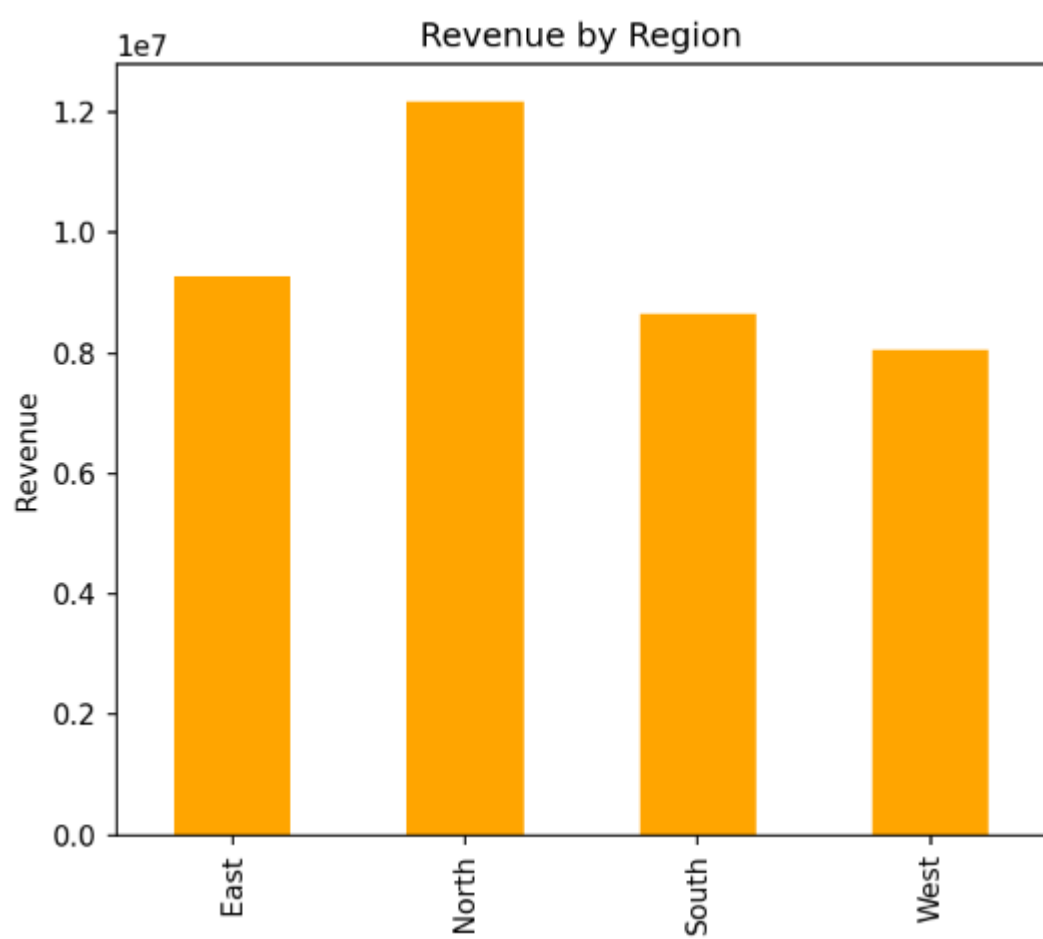
```
Statistics
      Quantity  ...      Revenue
count  438.000000  ...    438.000000
mean    4.025114  ...   87156.392694
std     2.024294  ...  121828.640178
min     1.000000  ...   1200.000000
25%     2.000000  ...   9000.000000
50%     4.000000  ...  24500.000000
75%     6.000000  ... 140000.000000
max     7.000000  ... 490000.000000
```

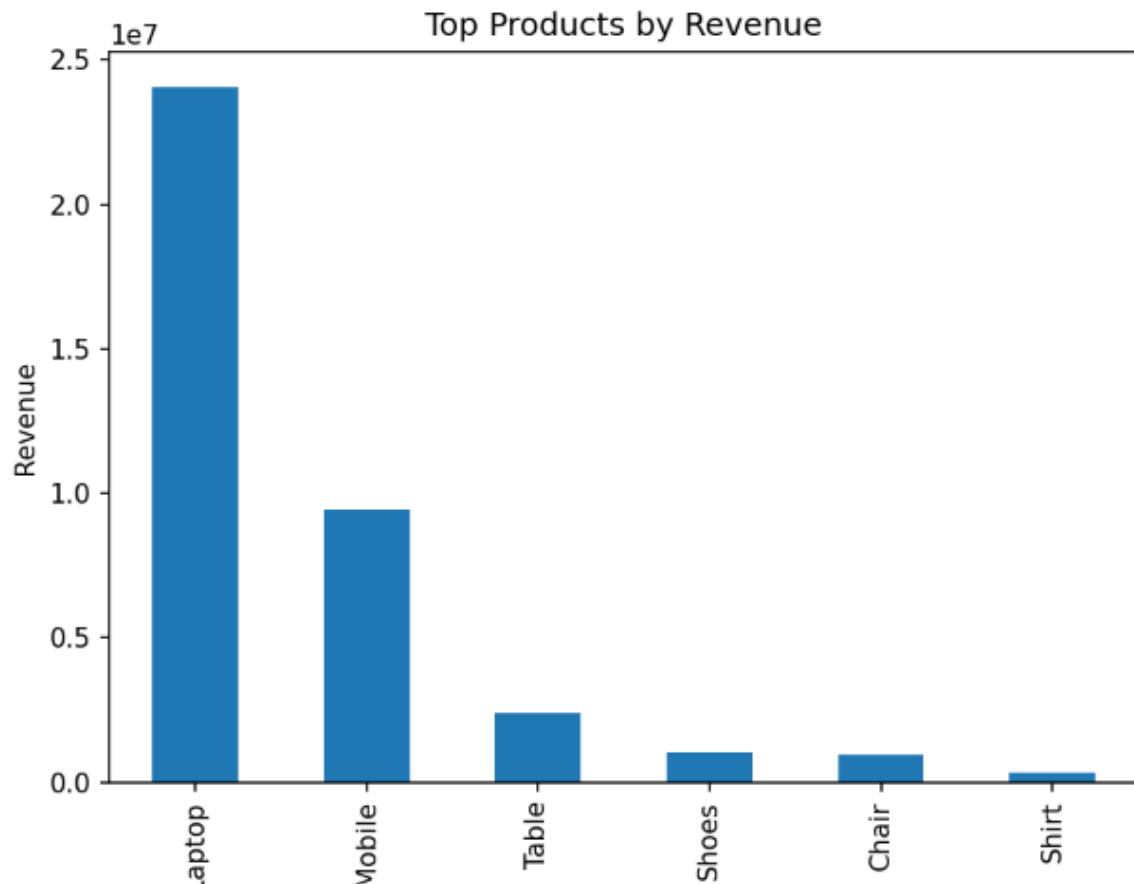
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Top Products
Product
Laptop    24080000
Mobile    9420000
Table     2376000
Shoes     1008000
Chair      948500
Shirt      342000
Name: Revenue, dtype: int64

Total Revenue : ₹ 38174500
Average Order Value : ₹ 87156.39
Best Category : Electronics
Best Region : North
```

```
Business Insights
- Highest revenue category : Electronics
- Highest revenue region : North
- Best selling product : Laptop
- Total Revenue : 38174500
- Average Revenue per Order : 87156.39
```







The retail sales dataset was successfully cleaned and analyzed using Python. Missing values in the dataset were identified and handled using the median (for Quantity) and mode (for Region). Duplicate records were removed to improve data quality before analysis.

Exploratory Data Analysis (EDA) was performed to study sales performance across different months, product categories, products, and regions. Revenue was calculated for each transaction using the formula:

$$\text{Revenue} = \text{Quantity} \times \text{Unit Price}$$

The analysis generated the following key results:

- The total revenue(38174500) generated from all sales transactions was successfully calculated.
- The average order value was determined to understand the average revenue per customer order.
- Monthly sales analysis identified the month with the highest revenue(Apr).
- Product-wise analysis identified the best-selling product based(Laptop) on total revenue.

- Category-wise analysis showed the product category contributing the highest revenue(Electronics).
- Region-wise analysis identified the highest-performing sales region(North).

The following visualizations were created:

- Monthly Revenue Trend (Line Chart)
- Revenue by Category (Bar Chart)
- Revenue by Region (Bar Chart)
- Top Products by Revenue (Bar Chart)

These visualizations helped in understanding sales patterns and identifying high-performing products, categories, and regions.

Overall, the project demonstrated the complete workflow of data cleaning, exploratory data analysis, data visualization, and extraction of meaningful business insights using Python.